



THE MAGIC GLOVE

Grade **06**

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Sustainable Development Goals (SDGs)
- Social-Emotional Learning (SEL)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- **SDG-Themed Modules** – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- **Inclusive & Engaging** – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

THE MAGIC GLOVE

Each week has a double session, total time of session 80-90 mins.



Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: When Words Don't Come.	2-21
Week 2		Lesson 2: Imagine, Sketch, Invent.	22-36
Week 3	Unit 2: Innovators in Action.	Lesson 1: Power of the Sense!	37-62
Week 4		Lesson 2: Glove on Paper.	63-80
Week 5		Lesson 3: Wired for Words.	81-101
Week 6		Lesson 4: Code the Conversation.	102-117
Week 7	Unit 3: Beyond Imagination.	Lesson 1: The Secret in Design.	118-135
Week 8		Lesson 2: Break Language Barriers.	136-155

About The Theme

What if a simple touch could transform silent gestures into readable messages—and even translate them into multiple languages in real time? In this theme, students become young inventors, coders, and problemsolvers as they design a smart glove that converts hand movements into digital text—helping people who use sign language communicate more easily with the world.

Inspired by real-life accessibility needs, students will explore how flex sensors, conductive materials, and embedded systems can detect finger gestures and translate them into readable language. These movements will be interpreted through an Arduino microcontroller and displayed as text on an LCD screen, turning physical motion into digital output and bridging gaps in communication.

Students will begin by investigating how flex sensors work—learning how bending these special materials changes their electrical resistance. Through Tinkercad simulations and coding exercises, they'll understand how the Arduino reads input from the sensors and controls visual output on an LCD. They'll then build a working prototype using copper tape, sensors, and coded logic, where each finger-touch gesture triggers a specific word or phrase—turning gestures into clear, readable messages.

To extend the impact, students will imagine and prototype the next evolution: connecting their glove to a mobile application that uses AI-powered translation tools. This step introduces them to the world of Natural Language Processing (NLP)—demonstrating how AI can translate text into different spoken languages, making communication more inclusive across both ability and language barriers.

Throughout the project, students will explore key interdisciplinary concepts, including:

- Sensor input and digital output.
- Arduino coding and logic.
- Empathy-driven, human-centered design.
- Artificial intelligence and real-time translation.

This theme aligns with Sustainable Development Goal 10 (Reduced Inequalities) and Goal 9 (Industry, Innovation, and Infrastructure), encouraging students to think globally and empathetically. By the end of the unit, they'll have built a smart, wearable translator—and discovered how AI, coding, and design thinking can be combined to create a more inclusive, connected world for everyone.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

Dig Deeper

Students will read articles that provide deeper insights into specific concepts, enhancing their understanding.

3

Problem-Based Learning:

- **Introducing the problem:**
Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- **Apply the Engineering Design Process (EDP):**
Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- **The Scientific Method:**
Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

B

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

5

Story

Join Adam and Laila, two fearless STEAM Scouts, on adventures full of discovery and creativity! From building inventions to solving real-world challenges, they'll inspire you to think, experiment, and innovate. Step into the world of science, technology, engineering, arts, and math—where every challenge becomes an adventure!

Story Reflection

In this section, students will reflect on the story's key themes, characters, and challenges.

6

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

7

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

8

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Understand how hearing and speech work, and why some people have trouble speaking or hearing.
- Show care and empathy for people who use different ways to communicate.
- Learn how assistive technology can help people express themselves.
- Explore how ancient civilizations like the Egyptians and Greeks used gestures to communicate.
- Use the engineering design process to solve a communication problem.
- Brainstorm creative, inclusive tech ideas that help people who can't speak or hear.
- Match hand gestures to symbols or words using simple logic and patterns.
- Understand how the brain sends signals to muscles to control hand movement.
- Learn how flex sensors work by changing resistance when they bend.
- Graph how resistance changes as the sensor bends, using paper or a digital tool.
- Design a circuit using flex sensors to read finger motion.
- Understand how copper tape carries electricity through a circuit.
- Sketch and plan your glove's circuit on paper before building.
- Explore how different cultures think about wearable technology and its uses.
- Check if a glove design is comfortable, clear, and useful for the person using it.
- Build a working smart glove using breadboards, sensors, and copper tape.
- Test your glove's wiring and fix any problems to make it work better.
- Write Arduino code that reads sensor data from the glove.
- Use "if-then" code to match finger movements to words or symbols.
- Add screens, apps, or visuals to your glove so others can read or hear the message.
- Connect your glove to a phone or device that shows or speaks the translated message.
- Learn how technology can help people around the world communicate across languages and abilities.

Unit

1

Lesson

1

Spark The Challenge!

When Words Don't Come.

Time Frame: [80-90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Explain how hearing and speech function in the human body.
- Identify possible causes of speech and hearing disabilities.
- Demonstrate empathy by discussing the challenges faced by individuals with communication differences.
- Describe examples of assistive technologies that support communication for people with disabilities.
- Investigate and compare gesture-based communication methods used in ancient civilizations, such as Egypt and Greece.

Challenge questions of the lesson

- How does the human body allow us to hear sounds and produce speech?
- What are some reasons why a person might have difficulty hearing or speaking?
- How do you think it feels for someone who cannot speak or hear to communicate with others?
- What kinds of technologies can help people who have trouble hearing or speaking?
- How did ancient civilizations like Egypt and Greece use gestures or symbols to communicate without speaking?

SEL Relation

- **Developing Empathy:** Students explore communication challenges faced by individuals with disabilities, encouraging understanding and compassion for diverse experiences.
- **Enhancing Social Awareness:** Through discussions and group activities, students learn to appreciate different communication styles and cultural methods, building respect for diversity.
- **Building Relationship Skills:** Collaborative teamwork during research and presentations strengthens cooperation, active listening, and effective communication among peers.
- **Promoting Self-Awareness and Self-Management:** Reflective activities encourage students to recognize their own communication styles and challenges, fostering self-expression and confidence.

Engage



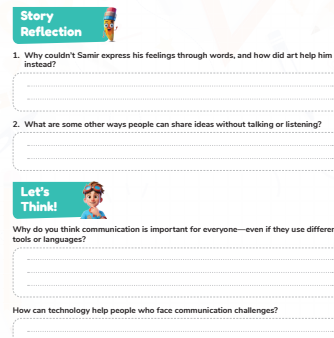
Teacher's Role:

- Read the story aloud with expression to engage students and bring the characters and setting to life.
- Pause at key points to check for understanding and invite students to imagine what the STEAM Scouts are experiencing.
- Guide students through the Story Reflection questions by encouraging thoughtful answers and connecting their ideas to the theme of communication without words.
- Facilitate the “Let’s Think” activity by asking the reflection questions aloud, prompting discussion, and encouraging students to consider how technology can assist people with communication challenges.
- Support diverse learners by offering examples or rephrasing questions to ensure all students understand and participate.

Expected From Students:

- Listen attentively to the story and visualize the scenes described.
- Reflect on the Story Reflection questions thoughtfully, sharing ideas about how Samir uses art to communicate.
- Actively participate in the “Let’s Think” discussion, expressing their own thoughts on the importance of communication and how technology can help.
- Respect others’ ideas during discussions and build on peers’ responses.

Parts of the book included.

Story Reflection

1. Why couldn’t Samir express his feelings through words, and how did art help him instead?
2. What are some other ways people can share ideas without talking or listening?

Let’s Think!

Why do you think communication is important for everyone—even if they use different tools or languages?

How can technology help people who face communication challenges?

Story



Teacher says (warmly, with curiosity):

- “Today, we’re going on a special visit to an art gallery with the STEAM Scouts. They meet Samir, an artist who shares his feelings in a unique way — through his sculptures instead of words. Let’s read and see how Samir’s art helps him communicate.”
- Then say: “As we read, listen closely to how Samir shares his thoughts without speaking, and how Adam and Laila try to understand him. Think about how it might feel to have so much to say, but not be able to use your voice.”

Possible Strategies for Implementation:

1. Read Aloud with Expression and Pauses.

- Use varied tone, pacing, and volume to bring characters and scenes to life.
- Pause periodically to ask predictive or reflective questions like, “What do you think Samir is feeling right now?” or “Why do you think Adam feels this way?”

2. Visualize the Setting and Characters.

- Encourage students to close their eyes briefly and imagine the art gallery and sculptures described.
- Ask them to describe what they picture to build visualization skills and deepen connection.

3. Use Think-Pair-Share.

- After reading key parts, have students turn to a partner and discuss their thoughts on Samir’s way of communicating or what they would feel in his place.
- Then invite volunteers to share with the whole class.

4. Relate to Prior Knowledge and Experiences.

- Prompt students to share times when they communicated without words (e.g., gestures, drawings, facial expressions).
- Connect their experiences to Samir’s use of art for communication.

5. Support Vocabulary Understanding.

- Pre-teach or pause to explain any challenging words (e.g., sculpture, communicate, feelings).
- Use images or gestures to support meaning.

6. Encourage Empathy and Perspective-Taking.

- Ask questions that help students imagine Samir’s feelings, such as “How do you think it feels to have a lot to say but not be able to speak?”
- Model empathetic language and responses.

7. Use Visual Supports

- Show pictures or examples of sculptures and other non-verbal art to help students connect with the concept.
- If available, use videos or short clips about artists who communicate through art.

8. Scaffold Story Reflection

- Guide students through the reflection questions with prompts or sentence starters (e.g., “I think Samir used sculpture because...”, “Another way people share ideas without talking is...”).
- Allow think time and encourage written or verbal responses.

9. Differentiate Participation

- Provide alternative ways to participate, such as drawing a response instead of speaking, or working in small groups.
- Offer additional support or scaffolding for students who need it.

Story Reflection



Objective:

Students will develop empathy by reflecting on how people express thoughts and feelings in different ways, including non-verbal communication, and critically consider the importance of alternative methods like art for sharing ideas when speaking is difficult.

Answers:

1. Why do you think Samir used sculpture to show his thoughts and feelings?

- Sample Answer:
 - “Samir used sculpture because he finds it hard to speak. Sculptures let him share his feelings and ideas without using words.”

2. What are some other ways people can share ideas without talking or listening?

- Sample Answer:
 - “People can use art, gestures, sign language, facial expressions, or writing to share ideas without talking.”

Possible Strategies for Implementation:

1. Modeling and Guided Discussion.

- Read each question aloud and model a thoughtful answer first.
- Encourage students to share their own answers verbally or in writing.
- Use sentence starters like “I think Samir used sculpture because...” or “Another way to share ideas is...”

2. Think-Pair-Share

- Allow students to think individually, then discuss answers with a partner before sharing with the whole class.
- This builds confidence and encourages diverse ideas.

3. Visual and Written Supports.

- Provide graphic organizers or reflection sheets with the questions and space to write or draw answers.
- Allow students who prefer drawing to express their ideas visually.

4. Encourage Multiple Modalities.

- Allow responses through speaking, writing, drawing, or even role-playing.
- This supports diverse learners and deepens understanding.

5. Use Real-Life Connections.

- Prompt students to think about times when they used non-verbal communication (e.g., gestures or drawings) to express themselves.
- Link their experiences back to Samir's story.

6. Check for Understanding and Clarify.

- Ask follow-up questions to probe deeper thinking, such as “Why is it important to find different ways to communicate?” or “How does using art help Samir?”

Let's Think!



Objective:

Students will critically think about the challenges of communication without words and explore how technology can provide solutions, developing problem-solving skills and empathy.

Answers:

1. Why do you think communication is important for everyone—even if they use different tools or languages?

Communication is important for everyone—regardless of the tools or languages they use—because it's the foundation of connection, understanding, and cooperation between people. Here's why it's essential:

- **Understanding and Empathy:** Communication helps us understand each other's feelings, thoughts, and needs. Even when we use different languages or tools (like sign language, speech devices, or gestures), being able to express and receive ideas builds empathy and respect.
- **Problem-Solving and Collaboration:** Working together to solve problems or reach goals relies on sharing ideas clearly. Different communication styles don't block this; they simply require more creativity or patience.

- **Inclusion and Equality:** Everyone deserves to be heard. Supporting diverse communication methods—like AAC (augmentative and alternative communication), braille, or different spoken languages—helps make society more inclusive.
- **Learning and Growth:** We learn from each other. Communication allows us to share knowledge, ask questions, and grow. If someone can't communicate, they may be left out of important learning opportunities.
- **Relationships and Belonging:** Relationships are built on communication—through words, signs, expressions, or even technology. When people can communicate in a way that works for them, they feel more connected and included.

2. How can technology help people who face communication challenges?

Technology can be a powerful tool for helping people who face communication challenges by giving them new ways to express themselves and connect with others. Here are some key ways technology helps:

1. Augmentative and Alternative Communication (AAC) Devices:

- These are tools that help people who can't speak or have limited speech. For example:
 - Speech-generating devices that turn text or symbols into spoken words.
 - Picture-based communication apps that let users select images to build sentences.

2. Text-to-Speech (TTS) and Speech-to-Text Tools:

- TTS tools help people who have difficulty speaking by reading aloud typed words.
- Speech-to-text tools help people who can't hear or process spoken language by converting speech into written text instantly.

3. Sign Language Apps and Video Chat:

- Apps can teach or translate sign language.
- Video chat platforms allow people to communicate visually using sign language or facial expressions.

4. Eye-Tracking and Head-Controlled Devices:

- These devices allow people with limited mobility to control a screen using only their eyes or head movements to communicate.

5. Hearing Aids and Cochlear Implants with Smart Features:

- These devices improve hearing and can connect to phones or apps, making conversations clearer and more accessible.

6. AI and Predictive Text:

- AI-powered tools can predict words or phrases to make typing faster for people with motor difficulties.

Possible Strategies for Implementation:

1. Facilitated Group Discussion

- Pose each question to the whole class and encourage a variety of answers.

- Use guiding prompts to expand ideas, such as “Can you think of other examples?” or “Why do you think that would help?”

2. Think-Pair-Share

- Give students time to think individually, discuss their answers with a partner, then share with the class.
- This supports quieter students and promotes deeper thinking.

3. Use Visual Aids

- Show images or short videos of communication devices or apps to inspire ideas.
- Encourage students to draw or write their invention ideas.

4. Encourage Creativity and Empathy

- Ask students to imagine themselves or a friend who needs help communicating, fostering empathy.
- Support innovative thinking by validating all ideas and encouraging “outside the box” solutions.

5. Sentence Starters and Graphic Organizers.

- Provide prompts like “I think communication is important because...” to support language development.
- Use simple graphic organizers to help students organize their thoughts.

6. Differentiate for Diverse Learners

- Allow responses through speaking, drawing, or acting out ideas.
- Pair students strategically to support those needing extra help.

Explorer's Toolkit



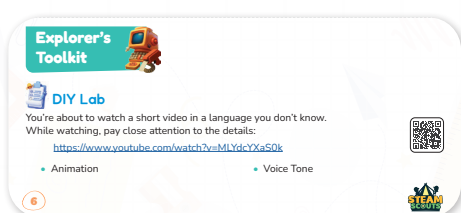
Teacher's Role:

- Prepare the classroom technology to play the video smoothly for all students.
- Introduce the activity by explaining that students will watch a video in a language they don't know and focus on how people communicate without words.
- Emphasize the importance of observing body language, facial expressions, and hand gestures to understand meaning.
- Guide students through each reflection question after watching, encouraging thoughtful responses and connections to the lesson's theme.
- Support students who might struggle with abstract observation by providing examples or rephrasing questions.
- Facilitate group or class discussions to share insights and build collective understanding.
- Encourage respectful listening and validation of diverse interpretations.

Expected from students:

- Watch the video attentively, focusing on non-verbal cues rather than the spoken language.
- Think critically about what the people in the video are trying to express through gestures and expressions.
- Respond thoughtfully to the reflection questions, sharing their observations and interpretations.
- Engage in discussions, respecting classmates' ideas and building on them.

Parts of the book included.



Don't worry about understanding the words—focus on how they communicate without speaking your language.

- What do you think the video was trying to say or express?

- What clues helped you understand—like movements, expressions, or voice tone?

- What was confusing or hard to figure out?

- How do you think people who use gestures or sign language feel when others don't understand them?



DIY Lab

Objective:

Students will develop observational and interpretive skills by analyzing non-verbal communication in a video, focusing on body language, facial expressions, and hand gestures to understand how people convey meaning without spoken words.

Video Link: <https://www.youtube.com/watch?v=MLYdcYXaS0k>



Answers

1. What do you think the person was trying to say or express?

- Sample Answer:
 - “The person was trying to show that they were happy, sad, confused, or excited through their facial expressions and body language.”

2. What clues helped you understand—like movements, expressions, or voice tone?

- Sample Answer:
 - “Smiling, waving hands, nodding, shaking the head, and the tone of voice helped me understand their feelings or message.”

3. What was confusing or hard to figure out?

- Sample Answer:
 - “Sometimes the gestures were unclear or I wasn't sure if the person was asking a question or telling a story.”

4. How do you think people who use gestures or sign language feel when others don't understand them?

- Sample Answer:
 - "They might feel frustrated, lonely, or sad because they want to share their thoughts but can't easily be understood."

Possible Strategies for Implementation:

1. Pre-viewing Preparation

- Briefly explain the purpose of the activity and key observation points (body language, facial expressions, gestures).
- Set clear expectations for attentive watching and respectful discussion.

2. Active Watching with Guided Focus.

- Play the video once for students to observe quietly.
- Optionally replay the video, pausing at key moments to highlight specific gestures or expressions.

3. Structured Reflection

- Present the reflection questions one by one, allowing students to write or discuss their answers.
- Use think-pair-share to support deeper thinking and verbalization.

4. Use Visual Supports

- Project key questions visibly during reflection.
- Provide graphic organizers or worksheets to record answers.

5. Encourage Empathy and Perspective-Taking.

- Prompt students to imagine how it feels to communicate this way and to be understood without words.
- Relate observations back to Samir's story and the overall theme of communication challenges.

6. Differentiate Support

- For students needing extra help, offer sentence starters or simplified questions.
- Allow responses through speaking, drawing, or other expressive means.

7. Link to Future Activities.

- Connect insights from the video to upcoming design challenges, highlighting how understanding non-verbal cues is essential for building communication tools.

Explain



Teacher's Role:

- Prepare the classroom for video viewing, ensuring all students can see and hear clearly.
- Introduce the video by explaining its focus on hearing and its importance in communication.
- Provide background context as needed to activate prior knowledge about human senses and technology.
- Guide students to take notes or mentally focus on key points during the video (e.g., parts of the ear, how sound is processed).
- Facilitate a group reading or individual reading of the article on assistive technologies, highlighting technologies related to communication challenges.
- Pause during reading to discuss key ideas, check understanding, and clarify any difficult vocabulary or concepts.
- Lead a whole-class discussion to connect the biological concepts with the assistive technologies, emphasizing how technology helps bridge communication gaps.
- Encourage questions and curiosity to deepen engagement.

Expected from students:

- Watch the video attentively, focusing on understanding how hearing works.
- Read or listen carefully to the article about assistive communication technologies.
- Take notes or discuss key concepts with peers or the teacher.
- Participate actively in discussions by asking questions and sharing insights.
- Make connections between the video, reading, and their prior knowledge or personal experiences.



Screen Lessons

Objective:

Students will gain a foundational understanding of the structure and function of the human ear and the biological process of hearing

Video link: <https://www.youtube.com/watch?v=mptjEoHF2aI>



Possible Strategies for Implementation:

1. Pre-Watching: Set a Purpose.

- **Strategy:** Activate prior knowledge about sound and introduce the concept of biological hearing.

- How to Implement:
 - Begin by asking: "How do we hear sounds? What parts of our body are involved when we listen to music or a friend talking?"
 - Introduce the video: "This video will explain how our ears work to allow us to hear."
 - Provide 1-2 guiding questions for students to think about while watching, such as: "What are the main parts of the ear?" or "How does sound travel through the ear to our brain?"

2. During Watching: Active Engagement.

- **Strategy:** Encourage active listening and note-taking on the parts of the ear and the sequence of hearing.
- How to Implement:
 - Play the video.
 - Pause and Explain: Briefly pause after each main section (e.g., outer ear, middle ear, inner ear) to allow students to summarize or ask questions about what was just explained.
 - "Key Terms" Jotting: Ask students to jot down any new technical terms they hear that describe parts of the ear or the hearing process (e.g., "pinna," "eardrum," "ossicles," "cochlea").
 - Sequential Notes: Encourage students to take notes on the step-by-step path of sound as it travels through the ear to the brain.

3. Post-Watching: Discussion and Reflection.

- **Strategy:** Debrief the biological concepts presented in the video and connect them to a broader understanding of sound perception.
- How to Implement:
 - **Quick Summary:** Ask students to briefly summarize (e.g., in pairs or as a class) the core idea of how sound is processed by the ear.
 - **Connect to Life:** Facilitate a discussion: "How does understanding how our ears work help us appreciate the sounds around us, or understand why some sounds can be damaging?"
 - **Q&A:** Open the floor for any questions students have about the biology of hearing. Address misconceptions and reinforce key concepts.
 - **Application Thought:** Ask: "How might a better understanding of human hearing influence the design of sound-related technology, like headphones, hearing aids, or even how we think about acoustics in building design?"



Dig Deeper

Objective:

Students will understand the definition, various types, and key benefits of assistive technology in facilitating communication, enhancing independence, and promoting inclusion for individuals with disabilities.

Activity link: <https://reciteme.com/news/assistive-technology-for-communication/>



Possible Strategies for implementation:

1. Pre-Reading: Activating Prior Knowledge & Setting Purpose.

- **Strategy:** Engage students by connecting to their existing knowledge about technology and challenges people might face, then set a clear purpose for reading.
- How to Implement:
 - Begin by asking students: "What kind of technology do you use to communicate every day? How might communication be challenging for some people?"
 - Introduce the concept of "assistive technology" and ask for any initial thoughts or examples.
 - Present the article title and explain that the purpose of reading is to learn about specific technologies designed to help people communicate more effectively. Ask guiding questions like: "What is assistive technology?" and "How does it help people with communication challenges?"

2. During Reading: Focused Information Gathering.

- **Strategy:** Guide students to actively identify and extract key information regarding types of assistive technology and their benefits.
- How to Implement:
 - Encourage students to highlight or take notes on the main definition of assistive technology provided in the article.
 - Instruct them to list specific examples of assistive communication devices or software mentioned (e.g., speech-generating devices, AAC apps, screen readers, hearing aids).
 - Have them identify and list the benefits of assistive technology for individuals with disabilities (e.g., improved communication, enhanced independence, better quality of life) and for broader society (e.g., diverse talent pools in employment).

3. Post-Reading: Consolidating Learning & Reflection.

- **Strategy:** Facilitate discussion and reflection to deepen understanding, clarify concepts, and connect the information to real-world applications.
- How to Implement:
 - **Quick Summary:** Ask students to briefly summarize the main types of assistive technology for communication they learned about and their primary purposes.
 - **Discussion:** Lead a discussion on how these technologies foster independence and inclusion. Ask: "Which examples of assistive technology surprised you the most?" or "How do these technologies support the idea of 'Universal Design'?"
 - **Connect to Our Project:** Prompt students to think about the "Smart Move" project: "How could our smart wheelchair project incorporate or be enhanced by some of the assistive communication technologies discussed in this article?"
 - **Reflection:** Ask students to reflect on the broader impact of assistive technology on society and individual lives.

Elaborate and Evaluate



Teacher's Role:

- Present the real-world communication challenges in each scene clearly.
- Encourage creativity by inviting students to write, draw, or act out solutions without limiting their ideas.
- Support students in imagining how their inventions might be used in daily life.
- Circulate to provide guidance, ask probing questions, and scaffold thinking.
- Demonstrate or explain the quick experiments on how the body makes sound.
- Facilitate the hands-on experiments, ensuring students perform them safely and correctly.
- Guide students in observing and recording their findings.
- Organize students into small teams for research on hearing or speech loss topics.
- Support collaboration, provide resources, and monitor progress to keep groups on task.
- Introduce the historical context of non-verbal communication methods in ancient civilizations.
- Guide students in researching common signs or symbols using books or the internet.
- Assist in organizing and creating clear, meaningful gesture cards.
- Encourage teamwork, creativity, and presentation skills as students share their findings.
- Facilitate reflection on how past communication methods relate to modern technology.

Expected from students:

- Observe each scenario carefully and think creatively about solutions to communication challenges.
- Express ideas through writing, drawing, or acting.
- Share their ideas confidently and listen respectfully to others.
- Participate actively in experiments, paying close attention to body sensations and sounds produced.
- Collaborate with team members to research hearing or speech loss topics thoroughly.
- Record observations and findings clearly.
- Engage respectfully in discussions, sharing insights and asking questions.
- Work collaboratively to research and understand ancient non-verbal communication.
- Create informative and visually clear gesture or symbol cards.
- Present their findings to peers, explaining the meanings and uses of the signs.

- Reflect on how communication has evolved and its importance.

Parts of the book included.



Apprentice

Objective:

Students will identify communication challenges faced by individuals with disabilities and creatively generate possible assistive technology solutions by expressing their ideas through writing, drawing, or acting.

Possible Strategies for implementation:

1. Contextual Introduction

- Begin by explaining the importance of communication and how disabilities can create unique challenges.
- Present the three illustrated scenes clearly, ensuring students understand the context of each situation.

2. Encourage Creative Expression

- Invite students to think beyond traditional solutions and encourage imaginative, diverse ideas.

- Offer multiple modes of expression—writing, drawing, or acting—to accommodate different learning styles and preferences.
- Use prompts such as “How would you help this person?” or “What technology could make communication easier here?”

3. Scaffold Idea Generation

- Provide sentence starters or question prompts for students needing extra support (e.g., “I would create a device that...”).
- Model a sample idea briefly to stimulate thinking without limiting creativity.

4. Foster Collaborative Sharing

- After individual or small group work, facilitate sharing sessions where students explain their ideas to peers.
- Encourage respectful listening and constructive feedback.

5. Link to Lesson Theme

- Relate student ideas back to the unit’s focus on assistive technology and communication tools.
- Highlight examples of real-world devices that solve similar challenges.

6. Differentiation

- Allow additional time or support for students with special needs.
- Pair or group students strategically to balance skills and encourage peer assistance.

7. Documentation and Reflection

- Encourage students to keep their sketches, writings, or videos as part of a project portfolio.
- Prompt reflection on how their solutions could impact people’s lives positively.



Builder

Objective:

Students will explore how the human body produces sound and understand challenges in speaking and hearing by conducting hands-on experiments and researching hearing and speech loss, developing scientific observation and critical thinking skills.

Activity instructions:

1. Part 1: How Does Your Body Make Sound?

- Preparation:
 - Distribute mirrors, straws, and cups of water.
 - Ensure each student has a clean straw and access to a safe space for experimenting.
- Steps:

1. Read the opening scenario aloud to spark curiosity and empathy:

- “Imagine trying to talk—but no sound comes out...”
- 2. Guide students through the three mini-experiments:
 - Humming while feeling their throat.
 - Watching their mouth movements in the mirror.
 - Blowing through a straw into water.
- 3. Encourage them to draw or write their observations in the provided space.
- 4. Use the illustrations as visual cues to demonstrate each step.
 - Tips:
 - Ask guiding questions:
 - “What do you feel?” “What do you see moving?” “What do the bubbles tell you?”
 - Emphasize how sound is made from vibration and airflow.

2. Part 2: What Happens When Hearing or Speech Is Lost?

- Preparation:
 - Provide 3 short resources (video clips, articles, or posters) related to:
 1. Hearing loss causes
 2. Voice box and speech disorders.
 3. Real people’s stories
- Steps:
 1. Split students into small teams (4–3 per group).
 2. Assign or let teams choose one topic.
 3. Teams explore their assigned resource and collect key ideas using the prompt on the page.
 4. Facilitate a short sharing session where each group presents what they learned.

CER Answers:

- **Claim:**
 - People can lose their hearing or ability to speak because of damage to parts of the ear or vocal cords.
- **Evidence:**
 - From the video and readings, I learned that loud noises, infections, or injuries can harm the ear and vocal cords. I also saw stories about people who lost hearing from illness or accidents.
- **Reasoning:**
 - Hearing and speaking depend on healthy ear structures and vocal cords. If these parts are damaged, sounds cannot be heard or produced correctly, which causes communication difficulties.

Possible Strategies for implementation:

Before the Activity (Pre-Learning).

1. Activate Prior Knowledge:

- Ask students:
 - “What do you know about how we speak or hear?”
 - “Have you ever lost your voice or had your ears blocked on a plane?”

2. Set a Purpose:

- Frame the activity with a question like:
 - “What makes communication possible—and what happens when it becomes difficult?”

3. Group Norms:

- Establish clear expectations for respectful communication, especially when discussing real stories or people with disabilities.

During the Activity

- **Part 1:** Hands-On Experiments.
 - Model each step briefly before letting students try.
 - Use Think-Pair-Share after each mini-experiment to compare observations.
 - Circulate to ask probing questions and clarify misunderstandings.
- **Part 2:** Research & Reflection.
 - Assign roles within teams: reader, recorder, presenter, timekeeper.
 - Use a Graphic Organizer or Guided Notes Template to help them capture key ideas.
 - Encourage use of visuals or sketches for better comprehension (e.g., voice box diagram, hearing aid image).

After the Activity (Consolidation & Reflection).

1. Group Presentations or Gallery Walk:

- Let students present findings or display posters summarizing their topic.
- Use sticky notes for peer feedback or questions.

2. Whole-Class Reflection Circle:

- Prompt discussion:
 - “What would life be like if we couldn’t hear or speak?”
 - “How can we design solutions for people who face these challenges?”



Mastermind

Objective:

- Students will explore how ancient civilizations communicated through gestures,

symbols, and hand signs. They will learn to recognize patterns and meanings of historical symbols and apply this knowledge to create their own visual symbols for modern communication.

Activity Instructions:

1. Introduction by Teacher (in class):

- The teacher will introduce the topic by discussing ancient communication methods such as the use of hand signs and symbols by civilizations like the Egyptians, Greeks, and Native Americans. They will guide students in exploring some historical examples and explain the importance of these visual languages for long-distance communication.

2. Students' Homework (at home):

1. Research a chosen ancient civilization (e.g., Egyptians, Greeks, or Native Americans) and identify symbols or hand signs they used for communication.
2. Using online resources, books, or AI tools like Preplixty, explore the meanings behind these symbols (e.g., "water," "help," or "friend").
3. Create flashcards with one symbol per card, writing the meaning of the symbol on the card (e.g., "water," "friend," etc.).
4. Draw or write these symbols on the cards and stick them in your notebook.



Now I Can

Objective:

Students will reflect on their learning and identify how they've grown in understanding communication, empathy, and the role of assistive tools in helping others share ideas.

Activity Instructions:

1. Introduce the Reflection

- Say: "Let's look back on everything we've explored this lesson. You learned how it feels to face communication challenges, and how tools can help people share their thoughts in special ways."

2. Guide Students Through the Checklist.

- Read each statement aloud one at a time.
- After each one, ask students to give a thumbs up, sideways, or down to show how confident they feel.
- Example prompts:
 - "Did you learn how people share ideas without talking?"
 - "Do you know how tools can help people talk?"

3. Student Completion

- Let students check off or color the "Now I Can" boxes they feel good about.

- Remind them: “It’s okay if you didn’t check all of them. We’re all still learning.”

4. Optional Extension

- Ask students to circle the one they’re most proud of.
- Invite them to write a sentence or draw a picture to show their favorite learning moment.

Relation to Standards

1. NGSS (Next Generation Science Standards).

- **MS-LS1-8:** Gather and synthesize information about how the body’s sensory systems detect and process information, including hearing and speech.
- **MS-ETS1-1:** Define problems that can be solved through engineering design, such as creating assistive technologies.
- **MS-ETS1-2:** Evaluate competing design solutions based on how well they meet criteria and constraints of communication challenges.

5. CCSS (Common Core State Standards).

- **CCSS.ELA-LITERACY.SL.5.1:** Engage effectively in collaborative discussions, building on others’ ideas and expressing their own clearly.
- **CCSS.ELA-LITERACY.RI.5.3:** Explain relationships or interactions between individuals, events, or ideas in texts, including scientific and technical texts.

6. ISTE (International Society for Technology in Education).

- **Empowered Learner:** Students leverage technology to seek knowledge and solve problems related to communication barriers.
- **Knowledge Constructor:** Students critically curate information from multiple sources on assistive technologies.